Personal information

Name / Surname	**Gelmetti, Ilario**
Address	Barcelona 08030, Spain
Personal Email	ilario@sindominio.net
Nationality	Italian
Date of birth	March 1988, Verona, Italy

Work experience

May 2023 – October 2025	**R&D Engineering Manager**. Vitsolc Technologies, Castelldefels, Barcelona, Spain (ICFO spin-off, Prof. Jordi Martorell group).
Project	Technology validation; **electronic engineering**; custom electronic setups design and fabrication; **coding in Python**; Linux server management; **projects writing and justification** (Eureka Eurostars, EIC Transition, CDTI Neotec, Acció Startup Capital); collaboration with ICFO researchers; MS Office suite; general chemistry lab.
July 2020 – May 2023	**Post-Doc researcher**. Prof. **J. R. Galán Mascarós** group, Institute of Chemical Research of Catalonia (**ICIQ**), Tarragona, Spain.
Project	Development of a spray coater, of a nitridation reactor and of large area electrodes for **heterogeneous electrocatalysis** for water splitting and for chemicals conversion (Horizon 2020 *DECADE* https://decadeproject.eu/), high pressure gas handling. **Project management**, justification and participation to international consortium meetings representing a whole work package. Design of experiments with machine learning.

Education

Jan. 2015 – July 2019	**Pre-Doc researcher**. Prof. **Emilio Palomares** group, Institute of Chemical Research of Catalonia (**ICIQ**) and Department of Electrical Electronic Engineering and Automation of Universitat Rovira i Virgili (DEEEA-URV), Tarragona, Spain. PhD evaluation excellent *cum laude* and international mention.
PhD Thesis Title	Advanced Characterization and Modelling of Charge Transfer in Perovskite Solar Cells.
Project	**Fabrication** (clean room, spin coating, glovebox, thermal evaporator) of perovskite solar cells. **Advanced characterization** using electrical and optical transients. Synthesis and characterization of materials, thin films and complete devices (FTIR, UV-vis, PL, TR-PL, DLS, XRD, NMR, TAS, AFM, SEM, EDX, optical microscopy, profilometer, EIS, solar simulator, oscilloscope); **Software development** for automated data analysis (R) and for data acquisition (Python). Inventory management.
Internship	3 months, Prof. **Jenny Nelson** Group, Dept. of Physics, **Imperial College London**, UK. Development of **numerical modeling** (Matlab) for simulation of impedance spectroscopy.
Oct. 2007 – Oct. 2013	Diploma of Science Class. Evaluation 70/70. **Scuola Normale Superiore di Pisa**, Italy.
Sept. 2010 – Oct. 2013	**Master Degree in Organic Chemistry**. Evaluation 110/110 *cum laude*. Dept. of Chemistry and Industrial Chemistry, **University of Pisa**, Italy. — Master Thesis: Synthesis and Characterization of Chiral Polythiophenes. — Laboratory: Institute for Macromolecular Studies, Consiglio Nazionale delle Ricerche, Milan, Italy.
Sept. 2007 – July 2010	**Bachelor of Science in Chemistry**. Evaluation 110/110 *cum laude*. Dept. of Chemistry and Industrial Chemistry, **University of Pisa**, Italy. — Bachelor Thesis: Synthesis of Conjugated Polymers and Quantum Dots for Photovoltaics Applications.
Sept. 2002 – June 2007	Secondary school diploma. Evaluation 100/100. ITIS G. Ferraris, Verona, Italy.

Volunteering

Creating Board Games	No Board Games (co-founder, 2014–). **Created the board games** *Cast the First Stone* (2015), *The Fully Automated Board Game* (2025) and *Attento al dvce* (2026).
Wireless Community Networks	eigenNet (co-founder, 2009–2013); LibreMesh.org modular firmware for community networks (**developer and community manager**, 2013–); NinuxVerona (founder, 2014–2015); organization of WirelessBattleMesh v15 (2023); GSoC project manager with Freifunk (written 3 successfully funded projects).
Hacklabs	eigenLab Pisa (co-founder, 2009–2013); TGNU Tarragona hacklab (co-founder, 2016–2018); Hacklab Logout (2022-2025); Phonelab Desendolla't (co-founder, 2026–).
Other volunteering	Operazione Mato Grosso (2004–2007), Manitese (2013).

Non-academic grants

June 2019 – August 2019 — **Google Summer of Code** (Freifunk and LibreMesh).
Project — Load-correlated distributed bandwidth analysis for LibreMesh networks
Acquired experience — Low level cabled and wireless networking; Lua coding; BASH scripting; git.

Awards

Chemistry Olympiads — **Fourth** position in category C, Italian **national level**, 2007.
Physics Olympiads — **Best experimental solution**, Italian **national level**, 2007.
Mathematics Olympiads — **Honorable mention**, Italian **national level**, 2006.

Personal skills

Mother tongue — **Italian**

Self-assessment
European level(*)

	Understanding			Speaking				Writing	
	Listening		Reading		Spoken interaction		Spoken production		
English	C1	Proficient user	C1	Proficient user	C1	Proficient user	C1	Proficient user	C1	Proficient user
Spanish	B2	Independent user	C1	Proficient user	C1	Proficient user	C1	Proficient user	B2	Independent user
Catalan	C1	Proficient user	C1	Proficient user	C1	Proficient user	C1	Proficient user	C1	Proficient user

(*) *Common European Framework of Reference* (CEF) level

Language certificates — Catalan language C1, issued by Universitat Rovira i Virgili, July 2020.
English language C1, issued by Escuela Oficial de Idiomas de Tarragona, February 2022.
Spanish language B2-C1, issued by SIELE, November 2025.

Computer skills — Linux system administration (more than 20 years experience using Debian and Arch Linux, x86 and ARM with DeviceTree); cabled and wireless networks setup and administration using OpenWrt; LaTeX; LibreOffice Writer, Calc and Impress; GIMP and Inkscape drawing; prototyping with Arduino and Single Board Computers for electronics (I2C, SPI, 1-wire, interrupt communications); assembling and repairing PCs and electronic hardware in general; CAD of mechanical pieces using FreeCAD (Part and Part Design workbenches).

Coding skills — Git version control; BASH scripting; advanced coding in Python; coding in R; coding in Matlab; coding in Lua.

Driving license — Driving license class B.

Online profiles

GitHub — https://github.com/ilario/

LinkedIn — https://www.linkedin.com/in/gelmetti/

ORCID — https://orcid.org/0000-0001-7115-1648

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References

Dr. **Sergi Cuesta Arcos**, CTO at Vitsolc Technologies, Castelldefels, Spain. sergi.cuesta@vitsolc.com

Prof. **José Ramón Galán-Mascarós**, Group leader at Institute of Chemical Research of Catalonia, Tarragona, Spain. jrgalan@iciq.es +34 977 920 200

Prof. **Emilio Palomares**, Director of Institute of Chemical Research of Catalonia, Tarragona, Spain. epalomares@iciq.es +34 977 920 245